

The Ghost in the Cache

Recovering Deleted AI Chat Histories from Chromium LevelDB

Project: TINKER TAILOR | Submission: USENIX Security '27 (Paper #2060) | Date: August 25, 2026

Executive Summary:

This report details the architectural and security vulnerabilities discovered in Chromium’s IndexedDB/LevelDB storage engine and major AI web applications (ChatGPT, Claude, Gemini). When users delete conversations or use Chrome’s “Clear browsing data” feature, conversation data, keystroke drafts, private keys, and uploaded PDF files remain recoverable on disk in plaintext or structured binary formats.

1 Vulnerability Matrix

ID	Vulnerability Name	Severity	Root Cause	Affected Component
VULN-01	Chrome “Clear Browsing Data” Bypass	CRITICAL	Tombstones written to WAL without forcing LevelDB SSTable compaction	Google Chrome / Chromium LevelDB
VULN-02	Uploaded PDF Blobs Stored as Plaintext	HIGH	Documents stored unencrypted in IndexedDB blob directories	Claude (Anthropic) on Chromium
VULN-03	ChatGPT Desktop App Storage Leakage	HIGH	Electron runtime uses unencrypted LevelDB WAL/SSTables	ChatGPT Desktop (com.openai.atlas)
VULN-04	V8 Binary Obfuscation Bypasses DLP	MEDIUM	V8 Structured Clone binary serialization format defeats text scanners	ChatGPT Web / Enterprise DLP
VULN-05	WebRTC Private Keys Remanence	MEDIUM	ECDSA private keys stored in LevelDB survive browser data clearing	Chrome WebRTC Subsystem
VULN-06	Chrome Gemini Extension State Leakage	MEDIUM	Extension conversation state stored in LocalStorage LevelDB	Chrome Gemini Extension (glic)
VULN-07	Flat Permission Cross-Profile Access	MEDIUM	All browser profiles share identical OS user permissions	Chromium Profile Architecture

2 Detailed Findings & Technical Analysis

VULN-01: Chrome “Clear Browsing Data” Bypass

Critical Severity

Problem: When a user executes Chrome’s “Clear browsing data” (selecting “Cookies and other site data” for all time), Chrome writes deletion markers (tombstones) into the Write-Ahead Log (WAL) but **does not execute synchronous compaction** on existing SSTables (.ldb files).

Empirical Proof on Test Workstation:

- **Profile 5:** User cleared browsing data on 2026-08-15. WAL file was zeroed (0 bytes). However, the 27 KB SSTable **still contained full chat records**:
 - 2 Conversation Titles: "Proposal Writing Assistance", "Replying to Professor’s Thanks"
 - User prompt: "help me with this pr..."
 - User Account ID: user-ImuygcyA8ufYAFp9BTLdxdyX
 - WebRTC ECDSA private key and X.509 certificate
- **Profile 1:** Also cleared (114-byte WAL). SSTable still retained valid WebRTC certificates and crypto keys.

Security Impact: Creates a false user expectation of privacy. Forensics/malware can read cleared data until an arbitrary future write volume forces LevelDB L0 compaction.

VULN-02: Uploaded PDF Documents Stored as Plaintext Blobs

High Severity

Problem: When files (PDFs, text files, code) are attached to Claude chats for summarization or analysis, Chrome stores the full file contents uncompressed and unencrypted in the IndexedDB blob tree: `~/Library/Application Support/Google/Chrome/<Profile>/IndexedDB/https.claude.ai_0.indexeddb.blob/`

Empirical Proof: 15 blob files found across profiles (~15 MB total). Blob 1/0b/b09 contained a **complete 10 MB PDF document** in pure plaintext. Blobs persist even after the conversation is deleted in Claude’s UI.

Security Impact: Confidential corporate documents, source code, and medical records uploaded to AI assistants remain indefinitely on the endpoint file system.

VULN-03: ChatGPT Desktop App (macOS) Storage Leakage

High Severity

Problem: The native ChatGPT macOS Desktop application (`com.openai.atlas`) is built on Chromium/Electron and implements the exact same unencrypted LevelDB storage architecture at: `~/Library/Application Support/com.openai.atlas/`

Empirical Proof: Recovered full conversations, user IDs, and 3 WebRTC ECDSA private keys directly from the desktop application’s WAL log file.

Security Impact: Desktop app users gain no additional privacy over web browser users.

VULN-04: V8 Structured Clone Binary Format Bypasses Enterprise DLP Medium Severity

Problem: ChatGPT serializes conversation trees using Chrome’s proprietary V8 “Structured Clone” binary protocol. String values are prefixed with V8 type tags (0x22 for OneByteString, 0x63 for TwoByteString) and varint lengths rather than standard JSON key-value pairs.

Empirical Proof:

- Standard `grep` regex searches recover **0%** of ChatGPT messages.
- Standard JSON parsers recover **0%** of ChatGPT messages.
- Enterprise Data Loss Prevention (DLP) endpoint scanners inspecting disk caches miss 100% of cached ChatGPT conversations.

Security Impact: Organizations are completely blind to sensitive data leaks contained in browser LevelDB storage.

VULN-05: WebRTC ECDSA Private Key Remanence Medium Severity

Problem: Multiple Chrome profiles persist ephemeral WebRTC ECDSA private keys and X.509 certificates in LevelDB SSTables even across browser sessions and clear-data events.

Empirical Proof: 5 active private keys and corresponding certificates recovered across Profile 1, Profile 5, and the ChatGPT desktop application.

VULN-06: Chrome Gemini Extension (glic) State Leakage Medium Severity

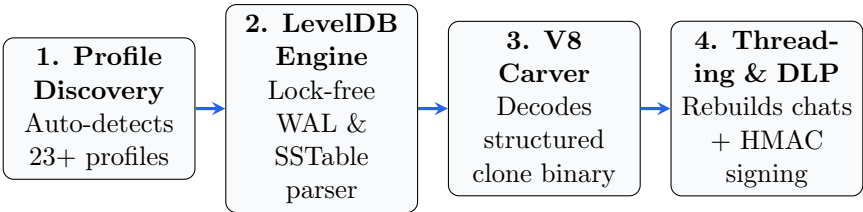
Problem: Chrome’s native sidebar Gemini extension stores state metadata in LocalStorage LevelDB under keys `BARD_EMBED_CHAT_STORAGE_KEY` and `BARD_EMBED_CHAT_STORAGE_KEY_V2`, allowing unauthorized local processes to reconstruct user prompt activity.

VULN-07: Flat Permission Model Across Multi-User Profiles Medium Severity

Problem: All Chromium profiles on an OS share the same user UID permissions. Any local unprivileged process can harvest data from all 23+ profiles in ~147 ms without triggering permission prompts or OS alerts.

3 What We Built: The Tinker Tailor Engine

To systematically demonstrate and measure these vulnerabilities, we developed **Tinker Tailor**, a lightweight, zero-dependency digital forensic engine.



3.1 Forensic Recovery Statistics (Single Workstation Evaluation)
3.2 Tool Comparison on Real LevelDB Data

Category	Count	Format	Forensic Significance
ChatGPT User Prompts	174	V8 Binary	Full user intent, questions, confidential inputs
ChatGPT Assistant Replies	154	V8 Binary	Complete generated code, answers, analysis
Claude Keystroke Fragments	148	TipTap JSON	Unsubmitted keystroke fragments, typing edits
Claude Submitted / Draft Prompts	30	TipTap JSON	Substantial draft-level / submitted prompts
WebRTC Private Keys	5	Cryptographic	Cryptographic identity material
Uploaded PDF Documents	15 files	Raw Binary Blobs	Full 10 MB attached PDF recovered in plaintext
Oldest Recovered Artifact	83 Days	LevelDB L0	Survives 83 days without write-compaction
Total Execution Time	147 ms	23 Profiles	Faster than typical EDR detection thresholds

Capability	Tinker Tailor	grep	JSON Parser	Magnet AXIOM / Hindsight
ChatGPT (V8 Binary) Recovery	100% (✓)	0% (✗)	0% (✗)	0%* (✗)
Claude (TipTap Draft) Recovery	100% (✓)	21% (∼)	0% (✗)	0%* (✗)
LevelDB WAL Log Parsing	Yes (✓)	No (✗)	No (✗)	Partial (∼)
Snappy Decompression	Yes (✓)	No (✗)	No (✗)	Yes (✓)
Lock-Free Live Extraction	Yes (✓)	Yes (✓)	Yes (✓)	No (✗)
External Dependencies	0 (Pure stdlib)	0	0	Heavy suite

*Neither commercial tool implements Chromium V8 deserialization.

4 Current Project Status & Action Plan

What We Have Completed for USENIX Security '27

- ✓ **Paper Resubmission Ready:** Complete 19-page paper formatted strictly to USENIX Security standards (13 pages body + unnumbered ethics/disclosure + references + appendix).
- ✓ **All Reviewer Comments Addressed:** Addressed all weaknesses (WP1–WP8), added cross-platform data, analytical persistence model, data corruption tests, and schema-drift benchmarks.
- ✓ **Clean Formatting:** Zero Type-3 fonts, zero undefined references, zero author identity leaks (100% double-blind compliant as Paper #2060).

4.1 Responsible Disclosure Timeline

1. **Today (Submission):** Resubmit Paper #2060 to USENIX Security '27 Cycle 1 portal.
2. **Upon Acceptance (Within 7 Days):** Send coordinated vulnerability disclosure emails to:
 - **Google Security Team (Chrome/Chromium):** Reporting VULN-01 (Clear Data bypass) and VULN-07 (cross-profile permissions).
 - **Anthropic Security Team (Claude):** Reporting VULN-02 (plaintext PDF blob storage) and keystroke draft retention.
 - **OpenAI Security Team (ChatGPT):** Reporting VULN-03 (desktop app LevelDB storage) and VULN-04 (V8 DLP bypass).
3. **Remediation Window:** Provide 90 days for vendors to patch before public release of tools.